

Table 4: Effects of AI adoption on wages

	(1) Wage (log) With controls	(2) Wage (log) With controls	(3) Wage/Empl.(log) With controls	(4) Wage/Empl.(log) With controls
AI adoption (actual)	0.135*** (0.010)		0.069*** (0.008)	
AI adoption (US-based)		0.024* (0.014)		0.033*** (0.011)
Investment (lag)	0.002 (0.004)	-0.000 (0.005)	0.005 (0.004)	0.002 (0.004)
Profitability (lag)	0.226*** (0.033)	0.269*** (0.033)	0.160*** (0.028)	0.193*** (0.028)
Financial leverage (lag)	-0.129*** (0.022)	-0.107*** (0.023)	-0.120*** (0.017)	-0.108*** (0.019)
Total assets (log, lag)	0.342*** (0.005)	0.354*** (0.005)	0.170*** (0.004)	0.171*** (0.004)
Size FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	34249	29304	34249	29304
R-squared	0.865	0.868	0.569	0.580
Adjusted R-squared	0.859	0.860	0.548	0.556

Note: The dependent variables are the log of the total wage bill (Columns 1-2) and the log of the average wage per employee (Column 3-4), and they are derived from EIBIS. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

Second, we find no evidence that AI adoption reduces employment in the short run, though longer-term effects remain uncertain. While our naïve specifications associate AI with 8% higher employment, the relationship vanishes after using predicted AI adoption. Moreover, our results suggest that on average the employees of AI-adopting firms benefit in terms of higher wages. This pattern alleviates near-term concerns about job destruction but raises questions about wage inequality. Indeed, benefits may accrue disproportionately to skilled workers coordinating AI-augmented workflows.¹¹

The divergence between significant productivity gains and null employment effects suggests that AI primarily enables capital deepening. This implies that firms increase worker output without reducing headcount, consistent with micro-level evidence that AI tools accelerate task completion without displacing labor (e.g., [Noy and Zhang \(2023\)](#)’s 40% time reduction in writing tasks; [Baily, Brynjolfsson, and Korinek \(2023\)](#)’s 14% productivity lift in customer support). To further support the capital deepening mechanism, we also test whether AI-adopting firms are more innovative. Table [B.4](#) in the Appendix presents the results, based on both actual adoption and our instrumented version, including firm-level controls and the same fixed effects as before. AI-adopting firms are more likely to develop products, services, or processes that are new for the firm (columns 1-2), country (columns 3-4) or global frontier (columns 5-6).

Third, in Table [5](#) we break down the effects of AI adoption by firm size groups. This heterogeneity analysis reveals that AI’s productivity benefits are concentrated. Medium and large firms exhibit stronger gains, highlighting the role of scale in absorbing integration costs and accessing complementary assets (e.g., data infrastructure, technical talent). This is also consistent with the important role that midcaps play in Europe ([EIB, 2024](#)). The disparities point to risks of widening productivity gaps between firms and regions. This echoes the warnings in [Cazzaniga et al. \(2024\)](#) about AI exacerbating inequality.

¹¹We also test whether the impact on employment varies by country using various indicators from the OECD employment protection database. We find no heterogeneous effects for the EU. The results are available upon request.

Table 5: Effects of AI adoption on labor productivity by company size

	(1) LP (log) Micro (5-9)	(2) LP (log) Small (10-49)	(3) LP (log) Medium (50-249)	(4) LP (log) Large (250+)
AI adoption (US-based)	-0.059 (0.068)	-0.041 (0.046)	0.040* (0.024)	0.079*** (0.024)
Investment (lag)	0.010 (0.012)	0.009 (0.008)	0.004 (0.014)	0.025 (0.020)
Profitability (lag)	0.326*** (0.089)	0.658*** (0.080)	0.735*** (0.080)	0.608*** (0.092)
Financial leverage (lag)	-0.256*** (0.080)	-0.162*** (0.050)	-0.072 (0.046)	-0.190*** (0.049)
Total assets (log, lag)	0.468*** (0.020)	0.414*** (0.012)	0.383*** (0.010)	0.325*** (0.011)
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	3761	7859	11256	6340
R-squared	0.597	0.615	0.590	0.583
Adjusted R-squared	0.522	0.560	0.535	0.500

Note: The dependent variable is labor productivity, calculated as the log of turnover per employee, and is derived from EIBIS. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

Last but not least, we shed more light on firm-specific enablers that spur productivity gains from AI adoption. Table 6 reports the size of marginal benefits from AI adoption, conditional on the level of complementary investments in different tangible and intangible assets. Investment in buildings and infrastructure seem to multiply the benefits of AI adoption (column (1)). However, those gains are small in magnitude compared to firms that invest in software and data (column (4)) or training (column (5)). An extra percentage point spent on software and data increases the effect of AI adoption on labor productivity by around 2.4%, and an extra percentage point spent on training by 5.9%.

Table 6: Effects of AI adoption on productivity with complementary investment in tangible and intangible assets

	(1) LP (log) Buildings	(2) LP (log) Machinery	(3) LP (log) R&D	(4) LP (log) Software	(5) LP (log) Training	(6) LP (log) Org.
AI adoption x Inv. share	0.007*** (0.003)	0.000 (0.002)	-0.005 (0.005)	0.024** (0.010)	0.059** (0.025)	0.012 (0.012)
Inv. share	0.000 (0.001)	0.003*** (0.001)	0.001 (0.002)	0.010*** (0.004)	0.019** (0.008)	0.005 (0.004)
AI adoption (US-based)	0.032 (0.025)	0.045* (0.026)	0.066*** (0.025)	0.030 (0.025)	0.035 (0.025)	0.050** (0.024)
Investment (lag)	0.011** (0.006)	0.010* (0.006)	0.012** (0.006)	0.011** (0.006)	0.011** (0.006)	0.012** (0.006)
Profitability (lag)	0.629*** (0.043)	0.612*** (0.042)	0.634*** (0.043)	0.627*** (0.043)	0.632*** (0.042)	0.635*** (0.043)
Financial leverage (lag)	-0.149*** (0.027)	-0.159*** (0.027)	-0.146*** (0.027)	-0.149*** (0.027)	-0.147*** (0.027)	-0.147*** (0.027)
Total assets (log, lag)	0.391*** (0.007)	0.397*** (0.007)	0.390*** (0.007)	0.395*** (0.007)	0.395*** (0.007)	0.391*** (0.007)
Size FE	Yes	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	27593	27593	27593	27593	27593	27593
R-squared	0.566	0.567	0.566	0.567	0.567	0.566
Adjusted R-squared	0.540	0.541	0.540	0.541	0.541	0.540

Note: The dependent variable is labor productivity, calculated as the log of turnover per employee, and is derived from EIBIS. AI adoption is the AI implementation status derived from similar firms in the US. Investment shares are EIBIS-reported investment in (1) Land, business buildings and infrastructure, (2) Machinery and equipment, (3) Research and development (including the acquisition of intellectual property), (4) Software, data, IT networks and website activities, (5) Training of employees and (6) Organisation and business process improvements, scaled by total assets. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.